

20250422EB_MN (Glo, 92)

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2025年4月22日 星期二

Use 2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A) or BP for differentiation and experiment.

Since the cells looked differently and not that healthy, I used more Na Pyruvate or BP to differentiate cells in the middle of the experiment.

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年4月24日 星期四

T2 -> T3

2025年4月26日 星期六

T3 -> T4 (~14 mL for over-weekend)

2025年4月28日 星期一

T4 -> T5

2025年4月30日 星期三

T5 -> T6

2025年5月2日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年5月5日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info			^
	A	B	
1	cell	passage	
2	WT11	P16+3	

Table1			^
	A	B	
1	run1	volume (mL)	
2	trypsin	6	
3	HI-FBS/CTWM	12	
4	run2	*take some undissolved EB	
5	trypsin	1.5	
6	HI-FBS/CTWM	3	

Table3											/
	A	B	C	D	E	F	G	H	I	J	
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume		
2	1.64	98	5	8.2						without the second round	
3	1.48	99	6	8.88	A	24	225	4	608		
4					B,C	96	55	144	the rest		

plate A							^
	1	2	3	4	5	6	
A							
B							
C	vLH92 (new, 4uL in total)						
D							

plate B													^	
	1	2	3	4	5	6	7	8	9	10	11	12		
A				T8 with 1.5mM pyruvate after 5/10										
B														
C														
D														
E				T8 with 0.5 mm pyruvate since 5/8										
F														
G														
H														

plate C													^
	1	2	3	4	5	6	7	8	9	10	11	12	
A				BP with 1.5 mM pyruvate since 5/8									
B													
C													
D													
E													
F													
G													
H													

2025年5月7日 星期三

T7-> T7 if not added virus

2025年5月8日 星期四

T7 -> T8
change to different T8 media (plate B,C see above)

plate A2							^
	1	2	3	4	5	6	
A							
B							
C	T8 with full supp (0.5mM pyruvate)						
D	BP T8 (1.5mM pyruvate)						

2025年5月10日 星期六

1/2 T8 -> 1/2 T8
change to different T8 media (plate B,C see above)

plate A3							^
	1	2	3	4	5	6	
A							
B							
C	T8 with full supp (0.5mM pyruvate)		T8 with full supp (1.5mM pyruvate)				
D	BP T8 with 1.5mM pyruvate						

2025年5月12日 星期一

1/2 T8 -> 1/2 T8

2025年5月14日 星期三

1/2 T8 -> 1/2 T8
plate C (plate B died) : Cell Titer Glo with different glucose concentrations

prepare ATP calibration							
	A	B	C	D	E	F	G
1	A	B	C	D	E	F	G
2	100mM	1mM	1uM	300nM	100nM	30nM	10nM
3	dilution factor from the previous one	100	1000		10		10
4					from C		from E
5		1uL A in 99uL media	1uL B in 999uL	150uL in 350uL media	50uL C in 450uL media	150uL in 350uL media	50uL D in 450uL media

prepare media with different conc. of glucose					
	A	B	C	D	E
1	diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)
2	1100	20	55	2000	36.4
3	20	7.5	2.6666666667	600	225.0
4	20	2.5	8	1000	125.0
5	2.5	0.5	5	600	120.0
6	2.5	0.15	16.6666666667	600	36.0
7	2.5	0.05	50	600	12.0
8	2.5	0.015	166.6666666667	600	3.6

plate B layout (cell died)													^
	1	2	3	4	5	6	7	8	9	10	11	12	
A	blank			0	0.015	0.05	0.15	2.5	0.5	2.5	7.5	20	
B	blank with 1.5 mM pyruvate												
C	C												
D	D												
E	E			2.5	0	0.015	0.05	0.15	0.5	2.5	7.5	20	
F	F												
G													
H	G												

Table2				^
	A	B	C	
1		12:30	prepare Celltiter glo reagent	
2	-105	14:40	prepare media into a 96-well plate	
3	-75	14:50-15:00	add media to cells	
4	-23	15:50	prepare ATP standard and add samples	
5	-17	16:00	add celltiter glo reagent	
6	-12	16:05	put the plate into cytation 5 and start program to shake	
7	-10	16:07	wait	
8	0	16:17	measure	

**I used 8-well suction to remove media and almost all cells detached on the plate
Therefore, I used plate C instead. I used multichannel pipette to remove and add media gently. The cells survived.

plate C layout												
	1	2	3	4	5	6	7	8	9	10	11	12
A	blank			0	0	0.015	0.05	0.15	0.5	2.5	7.5	20
B	blank with 1.5 mM pyruvate											
C	C											
D	D											
E	E			2.5	0.015	0.05	2.5	0.15	0.5	2.5	7.5	20
F	F											
G	G											
H												

2025年5月15日 星期四

Plate A: imaging (vLH92) end point only

plate A1							^
	1	2	3	4	5	6	
A							
B							
C	0mM glu & 0.5mM pyruvate	0mM glu & 1.5mM pyruvate					
D	0mM glu & 0.5mM pyruvate	2.5mM glu & 0.5mM pyruvate					

NT: all wells before treatment

no-glu: 60 min after treatment (same set of cells as NT (fileID))

- a. C1: full T8 w/ 0mM glu & 0.5mM pyruvate (1 - MN; 2 - non-MN)
- b. C2: full T8 w/ 0mM glu & 1.5mM pyruvate (3 - MN; 4 - non-MN)
- c. D2: full T8 w/ 2.5mM glu & 0.5mM pyruvate (5 - MN; 6 - non-MN)
- d. D1: full T8 w/ 0mM glu & 0.5mM pyruvate (7 - MN; 8 - non-MN)

***the record of the treatment is disappear, so I am not a 100% sure about the treatment of D2 and D1. They might be reversed.

Analysis: combine NT and no-glu